ENN581 Robot Motion, Control and Planning · Mock exam 2 — solutions

Model answers with an indicative marking scheme. Every number was verified by script (site/src/checks/mock.py).

Mock exam 2 indicative total 64 marks · 2 hours

Q1 · DH table of an RRP arm derive calculate 6 marks

A base joint rotates about the vertical z_0. At height H = 0.5 m, a second revolute joint with a horizontal axis tilts a boom away from the vertical by θ_2. The boom points straight *up* at $\theta_2 = 0$, and positive θ_2 tilts it outwards, in the direction set by θ_1. Both joint axes meet at one point. A prismatic joint extends the tool along the boom by d_3. (a) Give a proximal DH table [3]. (b) Find the tool position at $\theta_1 = 40°$, $\theta_2 = 30°$, $d_3 = 0.6$ m [3].

(a) Rows $(\alpha_{i-1}, a_{i-1}, \theta_i, d_i)$: $(0, 0, \theta_1, H)$, $(-90°, 0, \theta_2, 0)$, $(+90°, 0, 0, d_3)$. The axes intersect, so all a = 0 [3].

(b) Tip = $(c_1 d_3 s_2, s_1 d_3 s_2, H + d_3 c_2) = (0.2298, 0.1928, 1.0196)$ m [3].

Q2 · Inverse kinematics of the RRP arm derive calculate 6 marks

For the arm of Q1, derive the IK and evaluate it for p = (0.3, 0.4, 0.9) m. How many solutions are there?

$\theta_1 = \text{atan2}(y, x)$; $r = \sqrt{(x^2 + y^2)}$, $z' = z - H$; $d_3 = \sqrt{(r^2 + z'^2)}$; $\theta_2 = \text{atan2}(r, z')$ [3]. Numbers: $\theta_1 = 53.130°$, r = 0.5, z′ = 0.4, $d_3 = 0.6403$ m, $\theta_2 = 51.340°$ [2]. Second solution $(\theta_1 + 180°, -\theta_2, d_3)$; with $d_3 > 0$ there are exactly two [1].

Q3 · Four-bar: assembly modes calculate sketch explain 8 marks

A four-bar has ground $L_0 = 0.8$ m, input crank $L_1 = 0.3$ m (angle θ_1 at {S} = (0, 0)), coupler $L_2 = 0.7$ m, rocker $L_3 = 0.5$ m pivoted at (0.8, 0). (a) Find both positions of the coupler–rocker joint B at $\theta_1 = 60°$ and the coupler angle ϕ for each [5]. (b) Can the crank rotate fully? Justify [2]. (c) What is the mobility? [1]

(a) A = (0.15, 0.2598); |AD| = 0.7000 m. Intersect circle (A, 0.7) with circle (D, 0.5) [2]: B = (0.8075, 0.4999) with $\phi = 20.063°$, $\phi_1 = 89.138°$; or B = (0.4608, −0.3674) with $\phi = -63.636°$, $\phi_1 = -132.712°$ [3].

(b) Grashof: $s + l = 0.3 + 0.8 = 1.1 \le p + q = 0.5 + 0.7 = 1.2$, and the shortest link is the crank, so it turns fully (crank-rocker) [2].

(c) $3(4 - 4 - 1) + 4 = 1$ [1].

Q4 · Linear segment with parabolic blends calculate sketch 6 marks

A joint moves from 10° to 70° in 3 s using LSPB with blend acceleration 30 °/s². (a) Check that the acceleration is large enough [1]. (b) Find the blend time, the linear-segment speed, and the angle at the end of the first blend [3]. (c) Sketch θ, $\dot{\theta}$, $\ddot{\theta}$ against time [2].

(a) Need $\ddot{\theta} \geq 4\Delta/t_f^2 = 4(60)/9 = \mathbf{26.67}$ °/s²; 30 is enough [1].

(b) $t_b = \frac{t_f}{2} - \frac{\sqrt{\ddot{\theta}^2 t_f^2 - 4\ddot{\theta}\Delta}}{2\ddot{\theta}} = 1.5 - \frac{\sqrt{8100 - 7200}}{60} = \mathbf{1.0}$ s; speed $\ddot{\theta} t_b = \mathbf{30}$ °/s; θ(t_b) = 10 + ½(30)(1^2) = 25° [3].

(c) θ: parabola 10→25° (0–1 s), straight line 25→55° (1–2 s), parabola 55→70° (2–3 s). $\dot{\theta}$: ramp 0→30, flat 30, ramp 30→0. $\ddot{\theta}$: +30, 0, −30 (steps) [2].

Q5 · Jacobian of the W13 RRR arm derive explain 7 marks

For the RRR arm of Sample Q1 (position x = c_1r, y = s_1r, z = L_1 + L_2 + L_3s_2 + L_4s_{23}, with r = L_3c_2 + L_4c_{23}): (a) derive the 3×3 position Jacobian [3]; (b) show that det J = −L_3L_4 sin θ_3 · r and interpret each factor [3]; (c) evaluate det J for L_3 = 0.4, L_4 = 0.3 m at θ = (30°, 20°, 40°) [1].

(a) With $\rho = L_3 s_2 + L_4 s_{23}$: $\mathbf{J} = \begin{bmatrix} -s_1 r & -c_1\rho & -c_1 L_4 s_{23} \\ c_1 r & -s_1\rho & -s_1 L_4 s_{23} \\ 0 & r & L_4 c_{23} \end{bmatrix}$ [3].

(b) Expanding gives $-L_3 L_4 \sin\theta_3 \left(L_3 c_2 + L_4 c_{23}\right)$ [1]. sin θ_3 = 0: the elbow is straight or folded, at the workspace boundary [1]. r = 0: the wrist point lies on the base axis, so joint 1 cannot move it [1].

(c) r = 0.5259 m, det J = −0.12 sin 40° × 0.5259 = −0.04056 [1].

Q6 · Two-link arm with uniform rods derive calculate 8 marks

A planar 2R arm in a vertical plane (q_1 measured from the horizontal, q_2 relative to link 1, gravity along −y) has uniform rods: m_1 = 3 kg, l_1 = 0.6 m; m_2 = 2 kg, l_2 = 0.4 m (COM at mid-length, $I_c = ml^2/12$). (a) Write M(q) and g(q) [5]. (b) The arm is momentarily at rest at q = (45°, −30°) with $\ddot{\mathbf{q}}$ = (1, 2) rad/s². Find the joint torques [3].

(a) $M_{11} = I_1 + m_1\frac{l_1^2}{4} + I_2 + m_2\left(l_1^2 + \frac{l_2^2}{4} + l_1 l_2 c_2\right)$, $M_{12} = I_2 + m_2\left(\frac{l_2^2}{4} + \frac{l_1 l_2}{2} c_2\right)$, $M_{22} = I_2 + m_2\frac{l_2^2}{4}$ [3]; $g_1 = \left(m_1\frac{l_1}{2} + m_2 l_1\right) g c_1 + m_2\frac{l_2}{2} g c_{12}$, $g_2 = m_2\frac{l_2}{2} g c_{12}$ [2].

(b) M = [[1.6024, 0.3145], [0.3145, 0.1067]] kg·m², g = (18.3574, 3.7903) N·m; at rest C = 0, so τ = M$\ddot{\mathbf{q}}$ + g = (20.5888, 4.3181) N·m [3].

Q7 · PID gains and stability calculate explain 7 marks

A partitioned joint uses $f' = \ddot{x}_d + k_v\dot{E} + k_p E + k_i \int E\,dt$. (a) With k_p = 49 s^{-2}, choose k_v for critical damping of the PD part [1]. (b) Derive the third-order error equation and the

characteristic polynomial [3]. (c) For which k_i is the loop stable? Is $k_i = 100$ stable? [2] (d) What is the steady-state error under a constant disturbance? [1]

(a) $k_v = 2\sqrt{49} = 14\ s^{-1}$ [1].

(b) $\ddot{E} + k_v\dot{E} + k_pE + k_i\int E = f_{dist}/m$; differentiate: $\dddot{E} + k_v\ddot{E} + k_p\dot{E} + k_iE = \dot{f}_{dist}/m$; polynomial $s^3 + 14s^2 + 49s + k_i$ [3].

(c) Routh–Hurwitz: stable for $0 < k_i < k_vk_p = 686$. With $k_i = 100$ the roots are −10.140 and −1.930 ± 2.477j: stable [2].

(d) Zero: at steady state $k_iE = 0$ [1].

Q8 · Mecanum (Swedish 45°) robot derive calculate 6 marks

Four wheels (r = 0.05 m) at (±0.20, ±0.15) m all drive along x_C (β = 0). Front-left and rear-right rollers have γ = −45°; front-right and rear-left have γ = +45°. (a) Using the generalised model, write the four wheel rows [3]. (b) Find the wheel speeds for pure sideways motion v = 0.2 m/s [2]. (c) Is the robot holonomic? [1]

(a) $r\dot{\varphi}_i = (u - \dot{\theta}y_i) + \tan\gamma_i\,(v + \dot{\theta}x_i)$. Rows (per m/s; divide by r = 0.05): FL (0.2, 0.15, −45°): [20, −20, −7]; FR (0.2, −0.15, +45°): [20, 20, 7]; RL (−0.2, 0.15, +45°): [20, 20, −7]; RR (−0.2, −0.15, −45°): [20, −20, 7] [3].

(b) $\dot{\boldsymbol{\varphi}}$ = (−4, 4, 4, −4) rad/s: diagonal pairs turn the same way [2].

(c) The 4×3 matrix has rank 3, so any (u, v, $\dot{\theta}$) is achievable: holonomic [1].

Q9 · Planning concepts explain 6 marks

(a) Why are grid planners impractical for a 6-DoF arm? Quantify with 100 cells per joint [2]. (b) Compare PRM and RRT: what they build, when to use each, and their completeness [3]. (c) When is A* guaranteed optimal? [1]

(a) The node count grows exponentially with dimension: $100^6 = 10^{12}$ cells (curse of dimensionality) [2].

(b) PRM samples free configurations and connects neighbours into a roadmap, then searches it: multi-query, good for static environments. RRT grows a tree from the start towards random samples: single-query, handles differential constraints. Both are probabilistically complete (success probability tends to 1 as samples grow) and neither is optimal [3].

(c) With an admissible heuristic (h never overestimates). Consistency also lets the closed set stay closed [1].

Q10 · Mechanisms: actuation and G–K failure explain 4 marks

(a) Define fully actuated, under-actuated and (kinematically) redundant, with one example each [2]. (b) A planar body is hung from the ground by three equal parallel links. G–K gives M = 0, yet it moves. Explain [2].

(a) Fully actuated: actuators = DoF (6R arm with 6 motors). Under-actuated: fewer actuators than DoF (that arm with 4 motors; or 3-SPS from Mock 1). Redundant: more joint DoF than the task needs (7R arm for a 6-DoF pose) [2].

(b) N = 5, J = 6: $3(5 - 6 - 1) + 6 = 0$. G–K assumes independent constraints; with equal parallel links the third link repeats the others' constraint (one redundant constraint), so the true mobility is 1. Remove one link: $3(4 - 4 - 1) + 4 = 1$ [2].